

Technical Summary: HPC-Based Side-Channel Attack Detection

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1. Executive Summary

This project implements a micro-architectural security framework using Machine Learning to detect Side-Channel Attacks (SCA). By monitoring Hardware Performance Counters (HPC) at high resolution, we distinguish between benign computational tasks and malicious cache-contention attacks (Prime+Probe). The current milestone marks the successful transition from a single-OS baseline to a cross-environment universal detection model.

2. Environment Specifications

- **Primary Environment:** AWS EC2 t2.medium, Ubuntu 24.04 LTS.
- **Validation Environment:** Fedora 40 Server (Alternative Kernel).
- **Telemetry Tool:** Linux `perf` utility (Monitoring hardware registers).
- **Sampling Rate:** 50ms (Resolution selected for optimal attack-signal capture).

3. Development Phases & Achievements

Phase 1: Baselines & Benign Telemetry - Developed `victim.c`: A workload simulating standard process behavior. - Collected 1 hour of 'Benign' telemetry on Ubuntu (~73,000 samples). - Fixed 'Double Collector' race conditions where overlapping PIDs corrupted CSV files. **Phase 2: Attack Simulation** - Developed `attacker.c`: A high-frequency cache-eviction process targeting L1/L2 caches. - Collected 1 hour of 'Attack' telemetry on Ubuntu. - **Outcome:** Visual validation (EDA) confirmed that attacks produce 10x spikes in cache-misses relative to instructions. **Phase 3: Machine Learning Implementation** - Engineered 'Ratio Features' (Cache Miss Rate, IPC) to make the model architecture-agnostic. -

Trained a Random Forest Classifier achieving **99.98% accuracy** on the Ubuntu dataset.

4. Current Progress: Cross-OS Validation (Fedora)

We have successfully expanded the research to **Fedora 40** to test model portability. - **Challenges Overcome:** - Kernel Permissions: Adjusting ``perf_event_paranoid`` levels. - Package Management: Migrating from ``apt`` to ``dnf5``. - Data Normalization: Fedora's ``perf`` uses space-delimiters and thousand-separators (commas), unlike Ubuntu's standard CSV format. - **Current Status:** Collected 30 minutes of Benign and 30 minutes of Attack data on Fedora. - **Hybrid Loader:** Developed a robust Python ingestion engine that auto-detects OS-formatting quirks.

5. Technical Metrics (HPCs Monitored)

- **cache-misses:** Primary indicator of Prime+Probe eviction. - **cache-references:** Context for miss-rate calculation. - **instructions & cycles:** Measures process intensity/work done. - **branches & branch-misses:** Secondary telemetry for identifying complex attack logic.

6. Pending Tasks & Roadmap

- **Universal Modeling (In Progress):** Merging Ubuntu and Fedora data to create a single 'Distribution-Agnostic' model. - **Objective 5 (Next):** Implementation of a real-time detection script that triggers alerts when an attack starts. - **Generalization:** Testing against different CPU architectures (Intel vs. AMD) if hardware permits. - **Report Finalization:** Compiling findings into the final research paper.